

Graduate Microeconomics

Preferences, Utility, and Utility Maximization

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Outline

- 1 About this class
- 2 Why Do We Need a Utility Function?
- 3 Primitive Notions
- 4 Preferences
- 5 The Utility Function
- 6 Concavity, Quasi-concavity and Definiteness
- 7 The Utility Maximization Problem
- 8 First-Order Conditions
- 9 Common Utility Functions

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About this class

- 64 hours (32 + 32)
- Grading:
 - 1 Homework 20%
 - 2 Exam 80%
- Textbooks:
 - 1 Andreu Mas-Colell, Michael D. Whinston and Jerry R. Green, Microeconomic Theory, Oxford University Press, 1995
 - 2 Geoffrey A. Jehle and Philip J. Reny, Advanced Microeconomic Theory, Pearson Education Limited, 2011.
- Office hour: by appointment (online or in-person)

About me

- Fields of interest: Environmental, Public, Applied Micro
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Economics Studies How People Make Choices

Often associated with

- Interest rates, inflation, and economic growth
- Stock prices, firm profits, and market demand

But its object is much broader

- What consumers buy, and how much
- Whether individuals work, retire, marry, or have children
- Which transport mode commuters choose
- Whether citizens vote, evade taxes, or commit crime
- How firms set prices, hire workers, and invest

Across all of these settings, economics asks one question:

How do people make decisions?

Utility Maximization

Utility maximization

Almost everything in this course is built on those two words.

So before we use them, we owe ourselves two questions:

- 1 What *is* utility?
- 2 Why do we want a [utility function](#) at all?

Utility Is Not an Observable

Walk into a supermarket.

- Nothing in your head assigns 37 to the apples and 29 to the bananas.
- No price tag reports utility. No instrument reads it off a person.

And yet everything after this page is written in terms of $u(\cdot)$.
Utility is not observed. It is **constructed**.

**If we made it up,
why are we allowed to compute with it?**

Ordinality

Relabel utility with any strictly increasing f :

	u	$2u + 5$	u^3	$\sqrt{u}$
apples x	37	79	50,653	6.08
bananas y	29	63	24,389	5.39
Ranking	$x \succ y$	$x \succ y$	$x \succ y$	$x \succ y$

Every choice, every demand curve, every welfare number is identical.

But utility is not arbitrary

The **numbers** are ours: labels, with no empirical content.

The **ranking** is his: a hypothesis, and data can refute it.

The Primitive Is a Comparison

You did not score the apples. You put them in the cart instead of the bananas.

- What happens at the shelf is a **comparison**.
- “Better off” is a two-place word: better off *than* something.

The primitive is a relation, not a number.

Preference relation

$\succeq$ is a binary relation on X ; $x \succeq y$ means “ x is at least as good as y ”.

$$x \succ y \iff x \succeq y \text{ and not } y \succeq x, \quad x \sim y \iff x \succeq y \text{ and } y \succeq x.$$

From Relation to Function

$\succeq$ is a set of ordered pairs. That is all it is.

- You cannot differentiate a set of pairs.
- You cannot put it inside a Lagrangian.

But the consumer's problem is an optimization problem:

$$\max_{x \in X} u(x) \quad \text{subject to} \quad p \cdot x \leq w.$$

u **is not a claim about psychology.**
It is a change of language.

When that translation is available: next time.

Two Uses of a Utility Function

One: counterfactual analysis

What would people choose if prices, incomes or the options were different?

Two: welfare

Are they better or worse off — and by how much, in money?

The second does not follow from the first.

$$\text{choices} \dashrightarrow \succeq \longrightarrow \underbrace{x(p, m)}_{\text{what they would do}} \longrightarrow \underbrace{CV, EV}_{\text{what it is worth}}$$

Only the first arrow is an inference. The rest are derivations.

Counterfactual Analysis

Data show what people chose at the prices they faced. Policy questions are about prices, incomes or choice sets that were never observed.

A utility function answers such questions directly. Demand is the solution to

$$x(p, m) = \arg \max \{ u(x) : p \cdot x \leq m \},$$

so to predict behaviour under a different p or m , solve the same problem again at the new values.

A good that does not yet exist ?

Welfare Measurement

A policy usually changes several outcomes at once: quantities consumed, out-of-pocket spending, exposure to risk, health. These are measured in different units, so they cannot be summed.

A utility function turns them into a single number. Measure the change by the money needed at the new prices to restore the original level of utility:

$$CV = e(p^1, u^0) - e(p^0, u^0),$$

where $e(p, u)$ is the smallest expenditure that attains utility u at prices p .

An estimated demand function is not enough for this calculation. It requires the preferences behind the demand.

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The Four Building Blocks

Every model of consumer choice is assembled from four parts:

- 1 the consumption set X ;
- 2 the feasible set $B \subseteq X$;
- 3 the preference relation $\succeq$;
- 4 the behavioural assumption.

The Consumption Set

The consumption set X collects every consumption plan the decision-maker can conceive, achievable or not. It is the domain of $\succeq$, and of u when a utility function exists: every axiom below carries an implicit “over X ”.

With n infinitely divisible goods in non-negative quantities, $x = (x_1, \dots, x_n)$ and $X = \mathbb{R}_+^n$, where

$$\mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0 \forall i\}, \quad \mathbb{R}_{++}^n = \{x \in \mathbb{R}^n : x_i > 0 \forall i\}.$$

Prices, income and availability appear nowhere here. They belong to the feasible set.

The Consumption Set

Two goods. With $n = 2$ the consumption set is the non-negative quadrant $X = \mathbb{R}_+^2$, and a typical bundle is $x = (x_1, x_2)$ with $x_1, x_2 \geq 0$. This is the setting of the classical indifference-curve diagrams we will draw throughout.

Discrete goods. If each quantity is required to be an integer, the consumption set becomes the lattice $X = \mathbb{Z}_+^2$. Housing markets, where houses come in units, and labour-supply decisions, where a job is full-time or part-time, are two natural applications.

The Consumption Set

Minimal requirements on X (JR, Assumption 1.1)

$X \subseteq \mathbb{R}_+^n$; X is closed; X is convex; $0 \in X$.

Convexity holds for $\mathbb{R}_+^n$ but fails for the lattice $\mathbb{Z}_+^2$, since the midpoint of two integer bundles need not itself be an integer bundle. It is convexity of X that later allows us to speak of convex preferences and of interior solutions.

Closedness holds for $\mathbb{R}_+^n$ but fails for its interior $\mathbb{R}_{++}^n$; it matters because the representation theorem argues through boundary points. Requiring $0 \in X$ says only that consuming nothing must be conceivable.

The Feasible Set and the Behavioural Assumption

The feasible set $B \subseteq X$ collects those plans that are not merely conceivable but realistically obtainable, once we have accounted for the practical, institutional and economic constraints the consumer faces. If X is about imagination, B is about circumstance.

Its precise form is a modelling decision. The budget set $B = \{x \in X : p \cdot x \leq y\}$ is the case we shall use most often.

The model is then closed by a **behavioural assumption**: the consumer selects an available alternative that is most preferred according to $\succeq$. The preference relation itself is the subject of the next section.

Closed Sets

Closedness was required of X , and it is the entire content of the continuity axiom. Work in $\mathbb{R}^n$ with the Euclidean distance, and write $B_\varepsilon(x) = \{y : \|y - x\| < \varepsilon\}$ for the open ball of radius ε about x .

Definition

S is **open** if every $x \in S$ has some $B_\varepsilon(x) \subseteq S$, and **closed** if its complement is open.

Equivalently, and this is the form we shall use: S is closed if and only if it contains all its limit points, that is, $x^k \in S$ and $x^k \rightarrow x$ imply $x \in S$.

Closed Sets

The interval $[0, 1]$ is closed and $(0, 1)$ is open, while $[0, 1)$ is neither. The orthant $\mathbb{R}_+^n$ is closed and its interior $\mathbb{R}_{++}^n$ is open; every finite set is closed; and $\mathbb{R}^n$ itself is both open and closed.

- The intersection of *any* collection of closed sets is closed, and so is the union of *finitely* many.
- Arbitrary unions need not be closed: $\bigcup_{k \geq 1} [\frac{1}{k}, 1] = (0, 1]$.

Compactness

A set $S \subseteq \mathbb{R}^n$ is **bounded** if it lies inside some ball, and **compact** if it is closed and bounded.

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The Preference Relation

The objectives of the decision-maker are summarised in a **preference relation** $\succeq$ on X , a binary relation allowing the comparison of pairs of alternatives. Read $x^1 \succeq x^2$ as “ x^1 is at least as good as x^2 ”.

From $\succeq$ two further relations on X are derived:

$$x^1 \succ x^2 \iff x^1 \succeq x^2 \text{ but not } x^2 \succeq x^1,$$

$$x^1 \sim x^2 \iff x^1 \succeq x^2 \text{ and } x^2 \succeq x^1,$$

read as “ x^1 is **preferred** to x^2 ” and “ x^1 is **indifferent** to x^2 ”.

The Preference Relation

In much of microeconomic theory preferences are assumed to be **rational**. The hypothesis is embodied in two assumptions.

Axiom 1: Completeness

For all $x^1, x^2 \in X$: $x^1 \succeq x^2$ or $x^2 \succeq x^1$ (or both).

Axiom 2: Transitivity

For all $x^1, x^2, x^3 \in X$: if $x^1 \succeq x^2$ and $x^2 \succeq x^3$, then $x^1 \succeq x^3$.

Jehle and Reny call a relation satisfying both simply a *preference relation* (Definition 1.1); we follow the more common usage, since we shall often need to speak of relations that fail one of the two.

Completeness

Think of Axiom 1 as a questionnaire with two rules:

- you must tick a box — “I cannot say” is not allowed;
- you may tick $x \succeq y$, or $y \succeq x$, or *both* (both means $x \sim y$);

The strength of the assumption should not be underestimated: it says that the work of evaluating every pair of alternatives, however remote from common experience, has already been done.

Plenty of binary relations fail it. Take “is the brother of”: if John and Bob are unrelated, **neither** relates to the other.

Transitivity

Transitivity goes to the heart of what is meant by rationality. It rules out a sequence of pairwise choices in which preferences cycle: an apple at least as good as a banana, a banana at least as good as an orange, and yet an orange preferred to an apple.

Like completeness, the assumption is demanding when the alternatives lie far from common experience. Unlike completeness, it is also more fundamental: substantial portions of economic theory would not survive if agents could not be assumed to hold transitive preferences.

Properties of $\succ$ and $\sim$

Proposition (MWG, Proposition 1.B.1)

If $\succeq$ is rational, then

- $\succ$ is **irreflexive** ($x \succ x$ never holds) and **transitive**;
- $\sim$ is **reflexive** ($x \sim x$ for all x), **transitive** and **symmetric**;
- if $x^1 \succ x^2 \succeq x^3$, then $x^1 \succ x^3$.

The two axioms therefore discipline not only $\succeq$ but the relations derived from it. Neither derived relation is complete, however: $\succ$ leaves indifferent pairs unranked, and $\sim$ leaves strictly ranked pairs unrelated.

Failures of Transitivity

A consumer cannot detect a difference smaller than one slice of pie:

$$x \sim y \iff |x - y| < 1, \quad x \succ y \iff x \geq y + 1.$$

Complete. And yet

$$1.5 \sim 0.8, \quad 0.8 \sim 0.3, \quad \text{but} \quad 1.5 \succ 0.3.$$

A perceptual threshold makes **indifference intransitive**. Shades of grey paint and milligrams of sugar do the same.

Failures of Transitivity

Transitivity can also fail through the way the alternatives are framed. A consumer is buying a stereo for \$125 and a calculator for \$15, and learns that the calculator is \$5 cheaper at a branch twenty minutes away.

x^1 : travel, and take the \$5 off the calculator;

x^2 : travel, and take the \$5 off the stereo;

x^3 : buy both items at the first store.

Far more respondents make the trip when the \$5 comes off the calculator than when it comes off the stereo: $x^1 \succ x^3$ and $x^3 \succ x^2$. Yet when the trip is unavoidable they do not care which item carries the rebate: $x^1 \sim x^2$, and transitivity is violated.

Kahneman and Tversky (1984). The saving is \$5 throughout; only the price it is set against changes.

The Limits of Rationality Alone

With Axioms 1 and 2 alone, indifference sets may be **thick**, may have **gaps**, may bend upward, and may be wildly irregular.

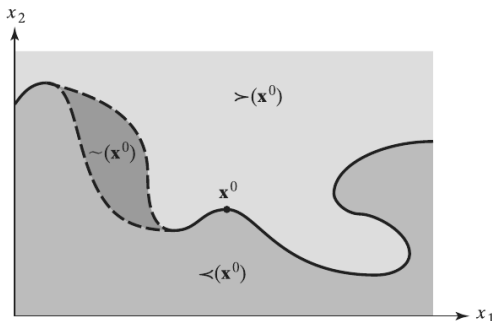


Fig. 1.1: preferences satisfying Axioms 1 and 2.

Nothing so far forbids any of it. Everything you remember drawing requires *more* axioms.

Continuity

For any $x \in X$ the **upper** and **lower contour sets** are

$\succeq(x) = \{y \in X : y \succeq x\}$ and $\preceq(x) = \{y \in X : x \succeq y\}$.

Axiom 3. Both are **closed** for every x : if $y^k \succeq x$ for all k and $y^k \rightarrow y$, then $y \succeq x$.

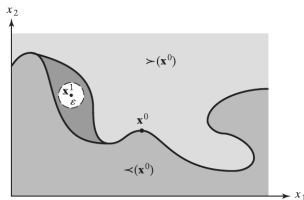


Fig. 1.2: Axioms 1, 2, and 3.

In Fig. 1.1 the indifference set was drawn *open* (dashed). Since $\sim(x) = \succeq(x) \cap \preceq(x)$ is an intersection of closed sets, it must now be **closed**. Continuity restricts limits, not tastes: the thick zone survives.

Continuity: lexicographic preference

Consider the **lexicographic** relation on $X = \mathbb{R}_+^2$, which ranks bundles by the first good and breaks ties by the second:

$$x \succeq_L y \iff x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2.$$

Take $x^0 = (1, 1)$ and the sequence $y^k = (1 - \frac{1}{k}, 2)$. Each y^k has first coordinate strictly below unity, so that $x^0 \succ_L y^k$ and hence $y^k \in \preceq(x^0)$ for every k .

The sequence converges to $(1, 2)$, and $(1, 2) \succ_L x^0$, since the first coordinates now agree while $2 > 1$. The limit therefore lies outside $\preceq(x^0)$, which is accordingly not closed, and $\succeq_L$ fails Axiom 3.

Local Non-satiation

Axiom 4'. For all x^0 and all $\varepsilon > 0$, there is some $x \in B_\varepsilon(x^0) \cap \mathbb{R}_+^n$ with $x \succ x^0$.

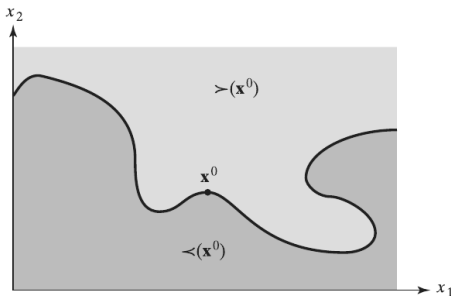


Fig. 1.3: Axioms 1, 2, 3, and 4'.

The zone of indifference is gone: no matter how small the ball, it must contain something strictly better. **No thick indifference sets.**

Monotonicity

Axiom 4. If $x^0 \geq x^1$ then $x^0 \succeq x^1$; if $x^0 \gg x^1$ then $x^0 \succ x^1$. More is better.

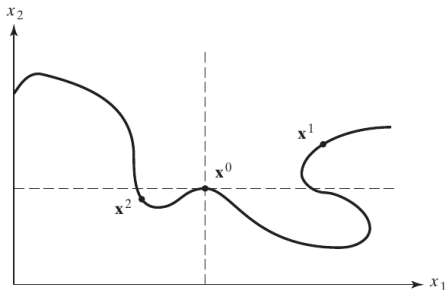


Fig. 1.4: why Axiom 4 bites.

Everything north-east of x^0 is strictly better, everything south-west strictly worse. So no indifference set may bend upward.

Monotonicity

Axiom 4 asks that *all* goods rise before we insist on strict improvement. A stronger version asks less:

$$x \geq y \text{ and } x \neq y \implies x \succ y.$$

One more unit of any *single* good already helps.

- $u(x) = x_1 + x_2$ satisfies both.
- $u(x) = \min\{x_1, x_2\}$ satisfies Axiom 4, but not the stronger version: from $(1, 1)$, more of good 1 alone changes nothing.

Non-convexity

Axioms 1–4 give the picture you remember: a thin, downward-sloping indifference curve, better to the north-east.

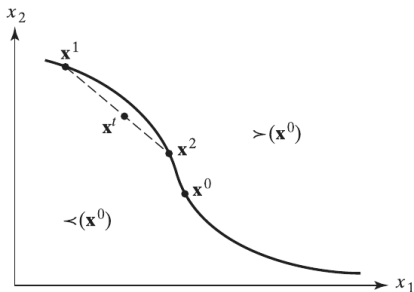


Fig. 1.5: Axioms 1, 2, 3, and 4.

Except in the north-west. There $x^1 \sim x^2$, yet the average x^t lies **below** the curve — the consumer is made *worse* off by balancing.

Convexity

5': $x^1 \succeq x^0 \Rightarrow tx^1 + (1-t)x^0 \succeq x^0$ for $t \in [0, 1]$. **5**: strictly, for $x^1 \neq x^0$ and $t \in (0, 1)$.

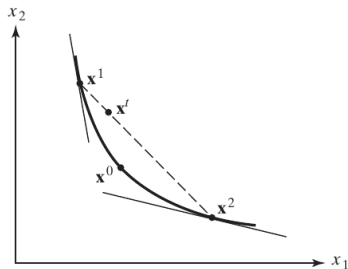


Fig. 1.6: Axioms 1, 2, 3, 4, and 5.

Balance beats extremes — equivalently, **diminishing MRS**: the rate at which the consumer gives up x_2 for x_1 does not rise as x_1 rises.

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Representing a Preference Relation

A utility function representing $\succeq$ (JR, Definition 1.5)

$u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ represents the preference relation $\succeq$ if, for all $x^0, x^1 \in \mathbb{R}_+^n$,

$$u(x^0) \geq u(x^1) \iff x^0 \succeq x^1.$$

A utility function assigns higher numbers to preferred bundles.
That is the whole of it.

“In modern theory, a utility function is simply a convenient device for summarising the information contained in the consumer’s preference relation — no more and no less.”

Representation on a Finite Set

If X is finite and $\succeq$ is rational, it is enough to count:

$$u(x) = \#\{y \in X : x \succeq y\}.$$

Transitivity makes the count rise along $\succeq$: everything below y is automatically below x . **Completeness** makes the converse work.

A menu: fish $\succ$ chicken $\succ$ vegetarian $\succ$ beef gives 4, 3, 2, 1. So does 100, 75, 50, 25.

On a finite set, rational = representable. No continuity needed.

The Representation Theorem

Existence of a representation (JR, Theorem 1.1)

If $\succeq$ is rational, continuous and strictly monotonic, there exists a **continuous** function $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ that represents $\succeq$.

- Rationality and continuity alone are already enough (Debreu 1954); monotonicity only shortens the proof.
- **No convexity required.** Representability says nothing about tastes.
- The three are also *necessary*, as we argued from $\mathbb{R}$.

An **existence** theorem only: at least one u exists, and nothing is said about its form.

Constructing the Utility Function

Let $e = (1, \dots, 1)$ be the bundle with one unit of every good.
Define $u(x)$ by the condition

$$u(x) e \sim x.$$

Assign to x the number $u(x)$ such that $u(x)$ units of *every* commodity is indifferent to x .

- Continuity guarantees such a number *exists*;
- strict monotonicity guarantees it is *unique*.

Uniqueness up to Increasing Transformations

Invariance to increasing transforms (JR, Theorem 1.2)

Let u represent $\succeq$. Then v also represents $\succeq$ if and only if $v(x) = f(u(x))$ for some f strictly increasing on the range of u .

So no significance whatsoever attaches to the numbers a utility function assigns — only to their **ordering**.

Properties of Utility Function

$\succsim$ is ...	if and only if u is ...
strictly monotonic	strictly increasing
convex	quasiconcave
strictly convex	strictly quasiconcave

We will discuss [concavity/quasi-concavity](#) in detail later.

Marginal Utility and the MRS

When u is differentiable, $\partial u(x)/\partial x_i$ is the **marginal utility** of good i , and

$$MRS_{12}(x) = \frac{\partial u(x)/\partial x_1}{\partial u(x)/\partial x_2}.$$

Replace u by $f(u)$: each marginal utility is multiplied by $f'(u)$, so the **ratio is unchanged**.

**Marginal utility is not observable.
The MRS is.**

That is why the marginal rate of substitution survived the ordinal revolution, and marginal utility did not.

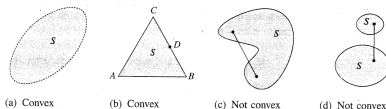
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Convex Sets

Definition

A set $S \subseteq \mathbb{R}^n$ is **convex** if $\theta x + (1 - \theta)y \in S$ for all $x, y \in S$ and all $\theta \in [0, 1]$.

Equivalently, the segment joining any two points of S lies entirely within S .



Convex and non-convex sets.

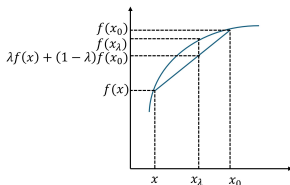
Concave Functions

Definition

A function f defined on a convex set S is **concave** if

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in S$ and all $\lambda \in (0, 1)$, and **convex** if the inequality is reversed.



A concave function: the chord lies below the graph.

Quasi-concave Functions

Two Definitions

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **quasi-concave** if

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$$

for all $x, y \in \mathbb{R}^n$ and all $\lambda \in [0, 1]$.

An equivalent formulation asks that every **upper level set** $S_\alpha = \{x \in \mathbb{R}^n : f(x) \geq \alpha\}$ be convex. The two statements are shown to agree on the next two slides.

This is the property that matters for us: quasi-concavity of u is precisely convexity of $\succeq$.

Quasi-concave Functions

The inequality implies convex upper level sets. Fix $\alpha \in \mathbb{R}$ and take $x, y \in S_\alpha$, so that $f(x) \geq \alpha$ and $f(y) \geq \alpha$.

For $\lambda \in [0, 1]$ write $z = \lambda x + (1 - \lambda)y$. The defining inequality gives

$$f(z) \geq \min\{f(x), f(y)\} \geq \alpha,$$

so that $z \in S_\alpha$.

Since x, y and λ were arbitrary, S_α is convex; and since α was arbitrary, every upper level set is convex.

Quasi-concave Functions

Convex upper level sets imply the inequality. Fix $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, and set $\alpha = \min\{f(x), f(y)\}$.

Then $f(x) \geq \alpha$ and $f(y) \geq \alpha$, so both points belong to S_α . By hypothesis S_α is convex, and therefore $z = \lambda x + (1 - \lambda)y \in S_\alpha$ as well.

Membership in S_α means precisely that

$$f(\lambda x + (1 - \lambda)y) \geq \alpha = \min\{f(x), f(y)\},$$

which is the inequality required.

Quasi-convex Functions

Definition

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **quasi-convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$$

for all $x, y \in \mathbb{R}^n$ and all $\lambda \in [0, 1]$.

Equivalently, every **lower level set** $\{x \in \mathbb{R}^n : f(x) \leq \alpha\}$ is convex; the argument is the one just given with the inequalities reversed.

The two notions are related in the obvious way: f is quasi-convex if and only if $-f$ is quasi-concave.

How to determine concavity/quasi-concavity?

The definitions state a condition on every pair of points and every λ , and are therefore difficult to verify directly for a given function.

For $n = 1$ this is not a serious obstacle: the graph can be drawn, and concavity on an interval is equivalent to $f''(x) \leq 0$. For $n = 2$ the graph is a surface, and for $n \geq 3$ it cannot be drawn.

For $n > 1$ the second derivative is not a number but the $n \times n$ Hessian. What is needed is a condition under which a matrix plays the role that a non-positive number plays when $n = 1$. That condition is **definiteness**.

Principal Minors

Let A be an $n \times n$ matrix.

Definition

A minor of order k is **principal** if it is obtained by deleting $n - k$ rows together with the $n - k$ columns bearing the same indices, and **leading** if the rows and columns deleted are the last $n - k$.

Deleting rows 1 and 3 therefore requires deleting columns 1 and 3 as well. We write D_k for the leading principal minor of order k , and Δ_k for any one of the $\binom{n}{k}$ principal minors of that order.

Principal Minors

For a 3×3 matrix $A = [a_{ij}]$ there are three principal minors of order 1, namely a_{11} , a_{22} and a_{33} ; three of order 2,

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, \quad \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix};$$

and one of order 3, the determinant of A itself.

Of these, the leading ones are $D_1 = a_{11}$, the first of the three determinants above, and $D_3 = |A|$. In general an $n \times n$ matrix has n leading principal minors and $2^n - 1$ principal minors.

The Hessian

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, the **Hessian** collects the second-order partial derivatives:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{bmatrix}$$

When the second derivatives are continuous $\mathbf{H}$ is symmetric, and the definiteness criteria that follow apply to it.

The Bordered Hessian

Bordering $\mathbf{H}$ with the first derivatives of f gives

$$H_B = \begin{bmatrix} 0 & \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \cdots & \frac{\partial f}{\partial x_n} \\ \frac{\partial f}{\partial x_1} & \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial f}{\partial x_2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_n} & \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

The first row and the first column hold the gradient; the remaining block is the Hessian itself. This is the matrix that governs quasi-concavity.

Definiteness

Theorem

Let $\mathbf{A}$ be a symmetric $n \times n$ matrix. Then $\mathbf{A}$ is

- **positive definite** if and only if $D_k > 0$ for every k ;
- **negative definite** if and only if $(-1)^k D_k > 0$ for every k ;
- **positive semidefinite** if and only if $\Delta_k \geq 0$ for every principal minor;
- **negative semidefinite** if and only if $(-1)^k \Delta_k \geq 0$ for every principal minor.

The two definite cases are settled by the n leading minors alone. The semidefinite cases are not: there all $2^n - 1$ principal minors must be checked.

Concave Functions

Theorem

Let f be C^2 on an open convex set S , with Hessian $\mathbf{H}$. Then f is convex on S if and only if $\mathbf{H}$ is positive semidefinite throughout S , and concave if and only if $\mathbf{H}$ is negative semidefinite throughout S .

Definiteness delivers the strict versions, but in one direction only: if $\mathbf{H}$ is positive definite on S then f is strictly convex, and if negative definite then strictly concave.

The converses fail. On $\mathbb{R}$ the function $f(x) = x^4$ is strictly convex, yet its second derivative vanishes at the origin.

Concave Functions

For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ of class C^2 the theorem reads: f is convex if and only if $f_{11} \geq 0$, $f_{22} \geq 0$ and $f_{11}f_{22} - f_{12}^2 \geq 0$.

Consider $f(x_1, x_2) = 2x_1 - x_2 - x_1^2 + x_1x_2 - x_2^2$, whose Hessian is

$$\mathbf{H} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}, \quad D_1 = -2, \quad D_2 = 3.$$

Since $(-1)^k D_k > 0$ for $k = 1, 2$, the Hessian is negative definite and f is strictly concave.

Concave Functions

Cobb–Douglas. On $S = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$ take $f(x_1, x_2) = x_1^a x_2^b$ with $a, b \geq 0$ and $a + b \leq 1$. Then

$$f_{11} = a(a-1)x_1^{a-2}x_2^b, \quad f_{22} = b(b-1)x_1^a x_2^{b-2}, \quad f_{12} = ab x_1^{a-1} x_2^{b-1},$$

the first two non-positive because $a, b \in [0, 1]$, while

$$f_{11}f_{22} - f_{12}^2 = ab(1-a-b)x_1^{2a-2}x_2^{2b-2} \geq 0.$$

All three conditions hold, so f is concave on S . If $a, b > 0$ and $a + b < 1$, then $f_{11} < 0$ and the determinant is strictly positive, so f is **strictly** concave.

The Bordered Hessian Criterion

Quasi-concavity can be read off the derivatives. For f of class C^2 , let $B_r(x)$ denote the leading principal minor of order $r + 1$ of the bordered Hessian,

$$B_r(x) = \begin{vmatrix} 0 & f_1(x) & f_2(x) & \cdots & f_r(x) \\ f_1(x) & f_{11}(x) & f_{12}(x) & \cdots & f_{1r}(x) \\ f_2(x) & f_{21}(x) & f_{22}(x) & \cdots & f_{2r}(x) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_r(x) & f_{r1}(x) & f_{r2}(x) & \cdots & f_{rr}(x) \end{vmatrix}.$$

If f is quasi-concave then $(-1)^r B_r(x) \geq 0$ for $r = 1, \dots, n$ and every x ; conversely, $(-1)^r B_r(x) > 0$ throughout is sufficient for [strict](#) quasi-concavity.

The Bordered Hessian Criterion

With $n = 2$ the criterion reduces to a single determinant,

$$B_2(x) = \begin{vmatrix} 0 & f_1 & f_2 \\ f_1 & f_{11} & f_{12} \\ f_2 & f_{21} & f_{22} \end{vmatrix} = 2f_1f_2f_{12} - f_2^2f_{11} - f_1^2f_{22}.$$

Since $(-1)^2 = 1$, quasi-concavity requires $B_2(x) \geq 0$; and $B_2(x) > 0$ together with $f_1(x) \neq 0$ throughout is sufficient for **strict** quasi-concavity.

The same expression appears when the marginal rate of substitution is differentiated along an indifference curve, which is why $B_2 \geq 0$ and a diminishing MRS are the same statement.

The Bordered Hessian Criterion: Example

Cobb–Douglas again. On $x_1, x_2 > 0$ let $z = Ax_1^{a_1}x_2^{a_2}$ with $A, a_1, a_2 > 0$. Then $z_i = a_i z / x_i$, $z_{ii} = a_i(a_i - 1)z / x_i^2$ and $z_{12} = a_1 a_2 z / (x_1 x_2)$, so that, after removing the common factor from each row and each column,

$$B_2 = \frac{a_1 a_2 z^3}{(x_1 x_2)^2} \begin{vmatrix} 0 & 1 & 1 \\ a_1 & a_1 - 1 & a_1 \\ a_2 & a_2 & a_2 - 1 \end{vmatrix} = \frac{a_1 a_2 (a_1 + a_2) z^3}{(x_1 x_2)^2} > 0.$$

Since $z_1 \neq 0$ as well, z is **strictly quasi-concave**, and this holds for *every* positive pair (a_1, a_2) .

Concavity required $a_1 + a_2 \leq 1$; quasi-concavity requires nothing. It is an ordinal property, and $x_1 x_2$ and $(x_1 x_2)^2$ rank bundles identically.

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The Utility Maximization Problem

Given prices $p \in \mathbb{R}_{++}^n$ and income $y > 0$, the consumer solves

$$\max_{x \in \mathbb{R}_+^n} u(x) \quad \text{subject to} \quad p \cdot x \leq y.$$

The solution $x(p, y)$ is the **Marshallian demand**, and the maximum value $v(p, y) = u(x(p, y))$ is the **indirect utility function**.

The axioms of the previous sections make u continuous, strongly monotone and strictly quasi-concave on $X = \mathbb{R}_+^n$. Each of the three does one job in what follows, and the next three slides identify which.

What Each Assumption Delivers

The axioms of the previous sections are not interchangeable. Each does one job in the problem that follows.

Assumption	What it delivers
Completeness, transitivity	an ordering; a best element if X is finite
Continuity	a continuous u ; with compactness, an optimum
Strict quasi-concavity	uniqueness on a convex feasible set
Local non-satiation	a binding budget constraint

Rationality orders the alternatives, continuity delivers existence, strict quasi-concavity delivers uniqueness, and local non-satiation makes the budget bind. The next three slides establish the last three in turn.

Existence

Proposition

If u is continuous on the budget set, the problem has a solution.

The budget set $B_{p,y} = \{x \in \mathbb{R}_+^n : p \cdot x \leq y\}$ is closed, and bounded because $p \gg 0$ forces $x_i \leq y/p_i$ for every i . It is therefore compact.

A continuous function on a non-empty compact set attains a maximum, by the Weierstrass theorem. Continuity of u is delivered by continuity of $\succeq$.

Uniqueness

Proposition

If u is strictly quasi-concave, the solution is a single point.

Suppose x^1 and x^2 both solve the problem, so that $u(x^1) = u(x^2) = v(p, y)$. The budget set is convex, being the intersection of a half-space with $\mathbb{R}_+^n$, so $z = \frac{1}{2}(x^1 + x^2)$ is feasible.

Strict quasi-concavity then gives

$u(z) > \min\{u(x^1), u(x^2)\} = v(p, y)$, contradicting the optimality of x^1 . Hence $x^1 = x^2$.

The Budget Constraint Binds

Proposition

If u is locally non-satiated and x solves the problem, then $p \cdot x = y$.

Suppose instead that $p \cdot x < y$, and set

$\varepsilon = (y - p \cdot x) / (2\|p\|) > 0$. The ball $B_\varepsilon(x)$ lies inside the budget set, and local non-satiation supplies a bundle x' in it with $x' \succ x$, contradicting optimality.

Strong monotonicity implies local non-satiation, so demand always exhausts the budget. This is the equation used to solve the problem.

The Role of Monotonicity

Strong monotonicity,

$$x \geq x', \quad x \neq x' \implies x \succ x',$$

implies local non-satiation, and therefore implies that the budget is exhausted at an optimum.

It is more than is needed for that conclusion. Local non-satiation alone suffices, and it does not require preferences to be increasing in every coordinate.

Monotonicity does no other job here. By itself it delivers neither uniqueness of the optimum, which requires strict quasi-concavity, nor a strictly positive multiplier.

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Interior Solutions

Assume in addition that u is C^1 . At an interior solution, where every $x_i > 0$, form the Lagrangian

$$\mathcal{L} = u(x) + \lambda (y - p \cdot x),$$

whose first-order conditions are

$$\frac{\partial u(x)}{\partial x_i} = \lambda p_i, \quad i = 1, \dots, n.$$

The multiplier λ is the same for every good. At the optimum, the marginal utility obtained per unit of expenditure is equal across goods.

The Tangency Condition

Taking the ratio of the first-order conditions for any two goods eliminates the multiplier:

$$MRS_{ik}(x) = \frac{\partial u(x)/\partial x_i}{\partial u(x)/\partial x_k} = \frac{p_i}{p_k}.$$

The rate at which the consumer is willing to substitute between two goods equals the rate at which the market permits the exchange. Geometrically, the indifference curve is tangent to the budget line.

These n conditions together with $p \cdot x = y$ are $n + 1$ equations in the $n + 1$ unknowns $x_1, \dots, x_n$ and λ .

Sufficiency of the First-Order Conditions

The conditions stated so far are necessary. That a bundle satisfying them is in fact optimal requires a separate argument, and quasi-concavity supplies it.

Theorem

Let u be continuous and quasi-concave on $\mathbb{R}_+^n$, with $(p, y) \gg 0$. If u is differentiable at x^* and $(x^*, \lambda) \gg 0$ satisfies

$$\nabla u(x^*) = \lambda p, \quad p \cdot x^* = y,$$

then x^* solves the utility maximization problem at (p, y) .

No second-order condition is required. Quasi-concavity is precisely the assumption that converts a local tangency into a global maximum.

Sufficiency of the First-Order Conditions

The proof uses one property of a differentiable quasi-concave function: for all $x, x^1 \geq 0$,

$$u(x^1) \geq u(x) \implies \nabla u(x) \cdot (x^1 - x) \geq 0.$$

Suppose x^* were not optimal. Some feasible x^0 then has $u(x^0) > u(x^*)$, and by continuity the slightly contracted bundle $x^1 = tx^0$, with t close to one, satisfies $u(x^1) > u(x^*)$ and $p \cdot x^1 < y$.

Using $\nabla u(x^*) = \lambda p$ and $p \cdot x^* = y$,

$$\nabla u(x^*) \cdot (x^1 - x^*) = \lambda (p \cdot x^1 - y) < 0,$$

which contradicts the property above. Hence x^* is optimal.

Sufficiency of the First-Order Conditions

When u is concave rather than merely quasi-concave, the argument is shorter. Concavity gives, for every feasible x ,

$$u(x) \leq u(x^*) + \nabla u(x^*) \cdot (x - x^*).$$

Substituting $\nabla u(x^*) = \lambda p$, and using $p \cdot x \leq y = p \cdot x^*$ together with $\lambda \geq 0$,

$$u(x) \leq u(x^*) + \lambda p \cdot (x - x^*) \leq u(x^*).$$

Concavity is therefore sufficient but not necessary.

Quasi-concavity, the ordinal property, already does the job at the cost of the slightly longer argument on the previous slide.

Corner Solutions

The tangency condition presumes an interior optimum. Whether one exists depends on how the marginal rate of substitution behaves near the axes.

- For Cobb–Douglas the MRS tends to infinity as a coordinate approaches zero, so the optimum is always interior.
- For perfect substitutes the MRS is constant. Unless it equals the price ratio, the optimum lies at a corner and the consumer spends her entire income on one good.

On the boundary the first-order condition holds only as an inequality. The Kuhn–Tucker conditions state the general case.

The Kuhn–Tucker Conditions

Attach a multiplier $\mu_i \geq 0$ to each constraint $x_i \geq 0$, and $\lambda \geq 0$ to the budget constraint:

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(y - p \cdot x) + \sum_{i=1}^n \mu_i x_i.$$

Necessary conditions

stationarity: $\partial u(x) / \partial x_i - \lambda p_i + \mu_i = 0$

primal feasibility: $p \cdot x \leq y, \quad x_i \geq 0$

dual feasibility: $\lambda \geq 0, \quad \mu_i \geq 0$

complementary slackness: $\lambda(y - p \cdot x) = 0, \quad \mu_i x_i = 0$

Interior and Boundary Goods

Stationarity combined with complementary slackness on $x_i \geq 0$ yields two cases:

$$x_i > 0 \implies \mu_i = 0 \implies \frac{\partial u(x)}{\partial x_i} = \lambda p_i,$$

$$x_i = 0 \implies \frac{\partial u(x)}{\partial x_i} \leq \lambda p_i.$$

Goods consumed in positive amounts satisfy the tangency condition of the previous slide. For a good that is not consumed, its marginal utility does not exceed the value of its price, so the first unit would not be worth buying.

The multiplier μ_i measures by how much the two sides differ.

A Worked Example

Let $u(x_1, x_2) = x_1 + 2\sqrt{x_2}$ with $p_1 = p_2 = 1$ and $y = 3$. Good 1 enters linearly, so the preferences are quasi-linear in it.

The first-order conditions are

$$\frac{\partial u}{\partial x_1} = 1 = \lambda p_1, \quad \frac{\partial u}{\partial x_2} = \frac{1}{\sqrt{x_2}} = \lambda p_2.$$

The first gives $\lambda = 1$, and the second then gives $x_2 = 1$. The budget equation delivers $x_1 = y - p_2 x_2 = 2$. Both goods are consumed, so $\mu_1 = \mu_2 = 0$ and the solution is interior.

A Worked Example

Now reduce income to $y = 0.5$. The interior conditions still give $x_2 = 1$, and hence $x_1 = 0.5 - 1 = -0.5 < 0$, which is infeasible.

The constraint $x_1 \geq 0$ therefore binds. Setting $x_1 = 0$ gives $x_2 = y/p_2 = 0.5$, and the conditions become

$$\frac{\partial u}{\partial x_1} = 1 \leq \lambda p_1, \quad \frac{\partial u}{\partial x_2} = \frac{1}{\sqrt{0.5}} \approx 1.414 = \lambda.$$

Since $1 \leq 1.414$, the multiplier is $\mu_1 = \lambda - 1 \approx 0.414 > 0$, and $\mu_1 x_1 = 0$ holds because $x_1 = 0$. The optimum is $x = (0, 0.5)$.

Reading the Multipliers

Dividing the first-order condition by p_i gives

$$\lambda = \frac{\partial u(x^*)/\partial x_i}{p_i} \quad \text{for every good with } x_i^* > 0.$$

The multiplier is the utility obtained from the last unit of expenditure, and it is the same whichever good that expenditure goes to.

At a corner, $\mu_i = \lambda p_i - \partial u(x^*)/\partial x_i \geq 0$ records how far short good i falls. It is the shadow value of the constraint $x_i \geq 0$.

Note that λ is not an ordinal magnitude: replacing u by $f(u)$ with $f' > 0$ replaces λ by $f'(u)\lambda$. Only ratios such as $MRS_{ik} = p_i/p_k$ survive.

The Envelope Theorem

Many objects in economics are maximum values that depend on a parameter:

$$M(a) = \max_x f(x, a), \quad x^*(a) = \arg \max_x f(x, a).$$

Envelope theorem, unconstrained

At an interior maximum, $dM(a)/da = \partial f(x^*(a), a)/\partial a$.

Differentiating $M(a) = f(x^*(a), a)$ gives

$$\frac{dM}{da} = \underbrace{\sum_i \frac{\partial f}{\partial x_i} \frac{dx_i^*}{da}}_{=0 \text{ at the optimum}} + \frac{\partial f}{\partial a}.$$

The indirect effect through the re-optimised choice vanishes to first order, precisely because the choice was already optimal.

The Envelope Theorem

With a constraint the same logic applies to the Lagrangian. Let

$$M(a) = \max_x f(x, a) \quad \text{subject to} \quad g(x, a) = 0,$$

$$\mathcal{L} = f(x, a) + \lambda g(x, a).$$

Envelope theorem, constrained

$dM(a)/da = \partial \mathcal{L} / \partial a$, evaluated at $(x^*(a), \lambda^*(a))$; that is,
 $\partial f / \partial a + \lambda^* \partial g / \partial a$.

Because $g(x^*(a), a) = 0$, we may write

$M(a) = \mathcal{L}(x^*(a), \lambda^*(a), a)$. Differentiating, the terms in dx^*/da and $d\lambda^*/da$ drop out, since $\partial \mathcal{L} / \partial x_i = 0$ and $\partial \mathcal{L} / \partial \lambda = g = 0$ at the optimum.

The Envelope Theorem

In the consumer's problem the parameter is income and the constraint is the budget, so $\mathcal{L} = u(x) + \lambda (y - p \cdot x)$.

Differentiating in y with the choice held fixed,

$$\frac{\partial \mathcal{L}}{\partial y} = \lambda, \quad \text{and therefore} \quad \frac{\partial v(p, y)}{\partial y} = \lambda.$$

This is the result obtained earlier by the chain rule, now in one line.

It also holds under weaker conditions. The earlier derivation differentiated the demand function $x(p, y)$; the envelope theorem never does, because the indirect effect is zero at the optimum.

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Cobb–Douglas Demand

The conditions above deliver closed forms. For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $\alpha \in (0, 1)$, the tangency condition reads

$$MRS_{12} = \frac{\alpha x_2}{(1 - \alpha)x_1} = \frac{p_1}{p_2}, \quad \text{so} \quad p_2 x_2 = \frac{1 - \alpha}{\alpha} p_1 x_1.$$

Substituting into $p_1 x_1 + p_2 x_2 = y$ gives

$$x_1(p, y) = \frac{\alpha y}{p_1}, \quad x_2(p, y) = \frac{(1 - \alpha)y}{p_2}.$$

Expenditure shares are constant at α and $1 - \alpha$, and each demand is proportional to income.

Functional Forms in Common Use

Cobb–Douglas	$u(x) = x_1^\alpha x_2^{1-\alpha}$
CES	$u(x) = (\alpha_1 x_1^\rho + \alpha_2 x_2^\rho)^{1/\rho}$
Perfect substitutes	$u(x) = \alpha_1 x_1 + \alpha_2 x_2$
Perfect complements	$u(x) = \min\{x_1, x_2\}$
Quasi-linear	$u(x) = x_1 + v(x_2)$

These five recur in nearly every application. The rest of this section asks which structural properties they share, and what each property implies for demand.

Homothetic Preferences

Definition

$\succsim$ on $X = \mathbb{R}_+^n$ is **homothetic** if, for all $x^1, x^2 \in X$ and all $\alpha > 0$,

$$x^1 \succsim x^2 \iff \alpha x^1 \succsim \alpha x^2.$$

Scaling both bundles by a common factor leaves the ranking unchanged, so the ordering depends on the composition of a bundle and not on its scale.

Geometrically, the entire indifference map is obtained by scaling any one indifference curve radially, and the *MRS* is constant along every ray through the origin.

Homothetic Preferences

Proposition

A continuous $\succeq$ on $\mathbb{R}_+^n$ is homothetic if and only if it admits a representation u that is homogeneous of degree one, that is, $u(\alpha x) = \alpha u(x)$ for every $\alpha > 0$.

One direction is immediate. If u is homogeneous of degree one, then $u(\alpha x^1) \geq u(\alpha x^2)$ holds exactly when $\alpha u(x^1) \geq \alpha u(x^2)$, that is, exactly when $u(x^1) \geq u(x^2)$.

Cobb–Douglas and CES are both homogeneous of degree one, and so are homothetic. Quasi-linear preferences are not.

Homothetic Preferences

Homotheticity restricts demand sharply. For any reference income $\bar{y}$,

$$x(p, y) = \frac{y}{\bar{y}} x(p, \bar{y}),$$

so demand is proportional to income and the income-expansion path is a ray through the origin.

Every good therefore has income elasticity equal to one. Cobb–Douglas illustrates the point: $\partial x_1 / \partial y = \alpha / p_1$, and dividing by $x_1 / y = \alpha / p_1$ gives exactly one.

This leaves no room for luxuries or necessities, which is why homotheticity is rejected in aggregate expenditure data.

Quasi-linear Preferences

Definition

$\succsim$ on $\mathbb{R} \times \mathbb{R}_+^{n-1}$ is **quasi-linear** in good 1 if it is represented by

$$u(x) = x_1 + v(x_2, \dots, x_n), \quad v \text{ continuous.}$$

Good 1 is the numeraire and enters linearly. For any $i \geq 2$,

$$MRS_{i1}(x) = \frac{\partial u / \partial x_i}{\partial u / \partial x_1} = \frac{\partial v}{\partial x_i},$$

which does not depend on x_1 . The indifference curves are parallel translates of one another in the numeraire direction.

Quasi-linear Preferences

Proposition

If v is strictly concave, the demands for goods $i \geq 2$ do not depend on income, provided the numeraire is consumed in a positive amount.

Normalise $p_1 = 1$ and substitute $x_1 = y - \sum_{i \geq 2} p_i x_i$ into the objective. The problem becomes

$$\max_{x_2, \dots, x_n} y - \sum_{i \geq 2} p_i x_i + v(x_2, \dots, x_n),$$

in which y appears only as an additive constant. The numeraire absorbs every income effect.

Quasi-linear Preferences

Because income effects on the other goods vanish, the three welfare measures coincide: $\Delta CS = CV = EV$. The area between the demand curve and the price line is then the exact utility change, measured in units of the numeraire.

With $u(x) = x_1 + \log(1 + x_2)$ and $p_1 = 1$, demand is $x_2(p_2) = \max\{0, 1/p_2 - 1\}$. A fall in p_2 from $\frac{1}{2}$ to $\frac{1}{4}$ raises x_2 from 1 to 3 and is worth

$$\int_{1/4}^{1/2} \left(\frac{1}{p} - 1 \right) dp = \log 2 - \frac{1}{4} \approx 0.44.$$

Away from quasi-linearity the area only approximates the welfare change, the error being the income effect.

Additively Separable Preferences

Definition

u on $\mathbb{R}_+^n$ is **additively separable** if $u(x) = \sum_{i=1}^n u_i(x_i)$ for some functions u_i .

Theorem (Debreu, 1960)

For $n \geq 3$, a continuous rational $\succeq$ admits an additively separable representation if and only if the ranking of two bundles that differ only in one subset of coordinates is independent of the levels held fixed in the remaining coordinates.

The trade-off between apples and oranges must not depend on how much coffee is consumed. This holds for goods far apart in the consumer's problem and fails for close complements.

The Three Properties Across the Families

	Homothetic	Quasi-linear	Separable
Cobb–Douglas	yes	no	in logs
CES	yes	no	in the ρ -th power
Perfect substitutes	yes	yes	yes
Quasi-linear $x_1 + v(x_2)$	no	yes	yes
Leontief	yes	no	no

Separability is the most demanding of the three. Cobb–Douglas and CES reveal it only after a monotone transformation, and Leontief has no separable representation at all: the trade-off between the goods depends on which of them is binding.

CES and the Elasticity of Substitution

For $u(x) = (\alpha_1 x_1^\rho + \alpha_2 x_2^\rho)^{1/\rho}$ the elasticity of substitution is constant,

$$\sigma = \frac{1}{1 - \rho},$$

and it governs the curvature of the indifference curves.

- $\rho \rightarrow 1$, hence $\sigma \rightarrow \infty$: perfect substitutes.
- $\rho \rightarrow 0$, hence $\sigma \rightarrow 1$: Cobb–Douglas.
- $\rho \rightarrow -\infty$, hence $\sigma \rightarrow 0$: Leontief.

Three of the five functional forms are therefore limiting cases of a single family, indexed by one parameter.